

1. Calculate and plot the value of  $y$  using solver *ode45* at all point for the interval  $t = 0$  to  $t = 2$  if

- (a)  $\frac{dy}{dt} = 12t^2 + 8t + 4$  with I.C.  $y(0) = 2$
- (b)  $\frac{dy}{dt} = y(e^t - 1)$  with I.C.  $y(0) = 1$
- (c)  $\frac{dy_1}{dt} = y_2 - 1$  and  $\frac{dy_2}{dt} = y_1 - t$  with I.C. are  $y_1(0) = 1$ ,  $y_2 = 1$
- (d)  $\frac{dy}{dt} = t - 0.1y^2$  with I.C.  $y(0) = 1$ , find the value of  $y(0.8)$
- (e)  $\frac{dy_1}{dt} = \frac{-1}{2}y_2 + \frac{y_1 y_2}{20}$  and  $\frac{dy_2}{dt} = y_1 + \frac{y_1 y_2}{10}$  with I.C. are  $y_1(0) = 1$ ,  $y_2 = 1$ .

2. Use the ode function defined in Practice Exercises 1 calculate and plot the value of  $y$  using *ode23* ode solver.

3. Calculate and plot the value of  $y$  using solver *ode45* and *ode23*, if  $A = \begin{bmatrix} -5 & 0 & -100 \\ 2 & -2 & 0 \\ 0 & 0.1 & -0.08 \end{bmatrix}$ ,  $B = \begin{bmatrix} 0 \\ 0 \\ -0.1 \end{bmatrix}$ , obtain the solution of the following set of differential equation for  $t = [0, 8]$

$$\frac{dy}{dt} = Ay + Bu \text{ with I.C., } y(0) = [0, 0, 0]'$$

where  $u = 0.2$ .

4. Solve the following differential equation exactly using *dsolve* command in Matlab:

- (a)  $\frac{dy}{dt} = ay$
- (b)  $\frac{dy}{dt} = a^2 y$  with  $y(0) = 1$
- (c)  $\frac{dy}{dt} = \sin(t)$
- (d)  $\frac{dy}{dt} = x$  and  $\frac{dx}{dt} = y$
- (e)  $\frac{dy}{dt} = x - 1$  and  $\frac{dx}{dt} = y - t$  with  $x(0) = 1$  and  $y(0) = 1$
- (f)  $\frac{dy}{dt} = y(e^t - 1)$  with  $y(0) = 1$ .